Cozy Bear Cottage （小熊中文） — Project Plan

Event: Meta Global AI Developer Hackathon (Meta Superintelligence Labs × Cerebral Valley) **Track:** Indie Developer · **Theme:** Build for Your World **Applicant:** Yukui McDonald (Indie track), building with the Bear team below **Bear team:** Patrick McDonald · Vivian McDonald · Evelyn McDonald · Yukui McDonald (team lead) · Xiaoyue He · Zhang Shun — the four adults (Yukui, Patrick, Xiaoyue He, Zhang Shun) are the registrable team (Indie track caps teams at 4, entrants 18+); Vivian is credited as the project's visual designer and Evelyn as part of the design team. **Key dates:** Application deadline Oct 25, 2026, 11:59 PM ET · Hackathon Oct 26 – Nov 4, 2026 · Submission due Nov 4, 9:00 PM ET · Finals Nov 9, 2026

Status (Oct 20, 2026): Project chosen: Cozy Bear Cottage. Application answers and packet prepared — **not yet submitted**; the only blank left is Yukui's GitHub link (LinkedIn added Oct 20). Pitch deck is 13 slides, designed around Vivian's artwork (see §10). The app skeleton is already built and verified (see §5), so the Oct 26 start is a head start, not a cold start.

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1. One-line pitch

An AI co-teacher for parent-run weekend heritage-language classes: it plans the lesson, illustrates the words, prints the worksheets, and turns any photo into a point-and-say Chinese game for kids ages 3–6.

2. The problem (firsthand)

Across the US, thousands of weekend Chinese schools are run by parent

volunteers with no curriculum team, no illustrator, and no budget. Every week the teacher hand-builds the same things from scratch: a 45-minute lesson plan, flashcards, printable worksheets, and games that hold a 3-to-6-year-old's attention. I know this problem firsthand — I run a Saturday Chinese class for young children in Bear, Delaware, and I prepare every one of these materials myself, every week.

Young children also learn language best from their own world — their cup, their dog, their drawing — not from generic stock flashcards. No existing tool connects a child's real surroundings to the week's vocabulary.

3. The product

Lesson Builder. The teacher picks an age range and a theme (colors, animals, numbers, family…). Muse Spark produces a complete 45-minute lesson: opening song, word set (characters + pinyin + English), a story, two games, and a bilingual teaching script the parent-teacher can follow word for word.

Living Flashcards. For each word in the lesson, Muse Image generates an illustration in one consistent, friendly style. One click exports a printable pack: flashcards, trace-the-character worksheets, and coloring pages (PDF, print-ready).

Point-and-Say （指着说）. The centerpiece. A child points a phone or tablet camera at real objects — or taps objects in any photo. SAM 3.1 segments the tapped object; Muse Spark names it in Chinese with pinyin, speaks it aloud, and invites the child to repeat it. A correct (or brave) attempt earns a star and a cheer. The week's target words are highlighted when the child finds them ("You found something 红色 — red!").

Parent view (stretch). Attendance plus a simple per-child record of

words encountered and mastered, so parents can practice the same words at home during the week.

Teaching dataset. Lesson themes and opening songs are grounded in a collected catalog of the "Chinese Playtime" sing-along series (Kristen Czyszek, with Lulu the Panda & Lenny the Koala; ~203K followers on Facebook, ~91.7K on Instagram): **111 song videos** (40 Facebook + 71 Instagram) deduplicated into **66 themed lessons**, each mapped to the words it teaches — greetings 你好・你好吗・我很好； counting 一 to 十; seven colours 红色・蓝色・黄色・绿色・橙色・紫色・粉红色； action words 跳・跑・走・坐・站・转・拍手； family 妈妈・爸爸・哥哥・姐姐・弟弟・妹妹・宝宝； transport 汽车・巴士・火车・自行车・船・飞机; emotions 快乐・难过・生气・害怕； daily routine 起床・刷牙・洗脸・上学. Every song maps to a lesson plan, its Living Flashcards, and Point-and-Say practice words for the Saturday class (ages 3–6).

4. How it uses Meta AI tooling (25% of judging)

- **Muse Spark** — lesson planning and teaching scripts; naming and explaining objects in context; generating practice sentences at a 3–6-year-old's level; adapting difficulty per child.
- **Muse Image** — the full flashcard and worksheet illustration set in one consistent character style (the little bear teacher).
- **SAM 3.1 (Meta Model API)** — tap-to-segment object recognition that powers Point-and-Say; this is the core interaction, not a decoration.
- **Muse Code** — used in the build itself.

5. Architecture

- Mobile-first web app (React or plain JS front end) — runs on the teacher's phone/tablet, no app-store step.
- Meta Model API for Muse Spark (text), Muse Image (illustrations),

and SAM 3.1 (segmentation).

- Speech: browser text-to-speech for Chinese audio; browser speech recognition for the repeat-after-me practice, scored leniently (encouragement-first for ages 3–6).
- No accounts or database in the MVP: class data lives in local storage; printable packs are generated PDFs.
- Hosting: any static + serverless deploy (e.g., Vercel/Cloudflare) with API keys server-side.
- **Already built (Oct 20, ahead of the window):** a React + Vite PWA skeleton with the three features stubbed (Lesson Builder, Living Flashcards, Point-and-Say), a serverless Meta Model API proxy, and the colors sample lesson wired end-to-end; `npm run build` verified. The hackathon days start from wiring real API calls into this skeleton, not from an empty repo.

6. Build plan, Oct 26 – Nov 4 (milestone-based)

1. **Oct 26–27:** Skeleton app + Meta Model API wiring; Lesson Builder end-to-end with one theme (colors).
2. **Oct 28–29:** Living Flashcards — Muse Image style lock-in, flashcard + worksheet PDF export.
3. **Oct 30–Nov 1:** Point-and-Say — SAM 3.1 tap segmentation, naming + audio loop, star rewards. (Highest-risk piece; buffer sits here.)
4. **Nov 2:** Test with real materials — next Saturday's actual lesson (colors) generated and printed.
5. **Nov 3:** Polish, offline-friendly fallbacks (pre-generated packs if the venue network fails), record the demo.
6. **Nov 4:** Final demo recording, project write-up, submit before 9:00 PM ET.

Cut line if time runs short: Parent view first, then per-child difficulty adaptation. The three core pieces (Lesson Builder, Flashcards, Point-

and-Say) are the submission.

7. Demo script (~3 minutes)

1. "I teach a Saturday Chinese class for 3–6-year-olds. This Saturday's theme is colors. Watch me prepare the whole class." Generate the colors lesson live.
2. Show the flashcard set and printable worksheets appearing from the same word list.
3. Point-and-Say: tap a red cup in a photo → segmentation outline → "红色， hóng sè — red!" → a child's voice repeats it → star.
4. Close: "Every weekend-school parent in America is doing this preparation by hand tonight. Cozy Bear Cottage does it in two minutes — and lets kids learn from their own world."

8. Judging alignment

- **Impact (30%)** — a real, widespread, underserved need: volunteer-run heritage schools and immigrant families working to keep a home language alive. Starts with one real class that meets every Saturday.
- **Use of Meta AI tooling (25%)** — all three flagship tools do essential work; remove any one and the product breaks.
- **Demo (25%)** — visual, joyful, live: a lesson appears, a worksheet prints, a child taps a cup and hears Chinese.
- **Technical depth (20%)** — multimodal pipeline (segmentation → language grounding → speech), age-adapted generation, print pipeline.

10. Visual design — by Vivian McDonald (age 4)

The project's visual identity is designed by Yukui's daughter Vivian. Her portfolio holds **48 original artworks** (paintings, coloring pages, paper

crafts — rainbow heart mandalas, a fairy ballerina, a Welcome sign, dotted flowers, rainbow mountains, and more), cut out as reusable art assets. The 13-slide pitch deck is built in her style (hand-painted rainbow stripe, light-blue ground, her cut-outs throughout — the Welcome sign on the cover), and the same assets carry into the app's look: flashcards, stickers, and reward art in a real child's hand, not stock illustration. Evelyn contributes to the design team.

11. Risks and mitigations

- **Speech recognition on toddler voices is unreliable** → score for participation and approximate pronunciation, not strict correctness; the star rewards trying.
- **Image style drift across a word set** → fixed style prompt + the same bear character as anchor; regenerate outliers.
- **Scope** → parent view is explicitly stretch; MVP is the three core pieces.

Application draft (project-idea section)

Project name: Cozy Bear Cottage （小熊中文）

What are you building? Cozy Bear Cottage is an AI co-teacher for parent-run weekend heritage-language classes, starting with Chinese for ages 3–6. A teacher picks a theme; Muse Spark generates the full 45-minute lesson plan and bilingual teaching script; Muse Image illustrates the week's vocabulary in one consistent style and exports printable flashcards and worksheets; and SAM 3.1 powers "Point-and-Say," where a child taps any object in a photo or through the camera, hears its Chinese name spoken, and repeats it for a reward. I run a real Saturday Chinese class for young children in Delaware and prepare all of these materials by hand every week — this tool is built from that firsthand

experience, for the thousands of volunteer parent-teachers doing the same.

Why does it matter? Heritage-language loss happens early: children who don't use their home language by age six often lose it. Weekend schools are the main defense, and they run on exhausted volunteers. Cutting weekly preparation from hours to minutes, and letting children learn words from objects in their own homes, directly serves those families.

Why is it feasible in the hackathon window? The MVP is a mobile web app on the Meta Model API — three well-separated features (lesson generation, image/printables, tap-to-segment naming), each demoable on its own. I am a senior machine learning software engineer; the riskiest piece (segmentation) is a single API capability, and the content pipeline is prompt engineering I can iterate quickly.

Applicant background (shared section): Senior Machine Learning Software Engineer at JPMorgan Chase (Nov 2018 – present), building AI/ML and platform systems in financial technology, with prior software development experience bringing total experience to ~12 years. Certifications: CKAD, AWS Certified Developer Associate, AWS Certified Solutions Architect Associate, HashiCorp Terraform Associate. PhD researcher at the University of Delaware (AI-driven cybersecurity for blockchain and digital currency systems). Also founder-teacher of a Saturday Chinese class for children ages 3–6 in Bear, Delaware. [Add: GitHub URL] · LinkedIn: https://www.linkedin.com/in/yukui-mcdonald-635739291